Course Title: Decision Theory BME 3
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Exam Session: Resit
Instructions: Answer All Questions

QUESTIONS

1. Expected utility theory assumes that individuals:
   A) Always avoid risk
   B) Maximize the sum of payoffs
   C) Maximize the expected value of utility across outcomes
   D) Choose randomly between options
2. A lottery in decision theory refers to:
   A) A game of luck in a casino
   B) A choice between two consumption bundles
   C) A probability distribution over outcomes
   D) A decision tree
3. If a person prefers the expected value of a lottery to the lottery itself, they are:
   A) Risk-loving
   B) Risk-averse
   C) Risk-neutral
   D) Irrational
4. A utility function that is concave indicates:
   A) Risk neutrality
   B) Risk aversion
   C) Risk loving
   D) Indifference
5. Suppose $U(x) = x$. An individual is:
   A) Risk-averse
   B) Risk-neutral
   C) Risk-loving
   D) Indifferent to income
6. The expected utility model is valid only when the decision maker's preferences satisfy:
   A) Linearity
   B) Budget constraint
   C) The von Neumann–Morgenstern axioms
   D) No uncertainty
7. The certainty equivalent is:
   A) The minimum utility level of a lottery
   B) The sure amount an individual values equally as the lottery
   C) The expected payoff of a risky investment
   D) The difference between utility and payoff
8. A risk-averse person's certainty equivalent is:
   A) Equal to the expected value
   B) Greater than the expected value
   C) Less than the expected value
   D) Equal to the risk premium
9. The risk premium is:
   A) The utility gain from taking risk
   B) The extra amount required to accept risk
   C) The cost of a good under uncertainty
   D) The expected value minus the certainty equivalent
10. A risk-neutral agent:
    A) Has a concave utility function

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B) Avoids all uncertain prospects
C) Is indifferent between a sure outcome and a risky one with the same expected value
D) Always chooses the risky option

11. For a lottery yielding 40 (0.5) and 90 (0.5), and $U(x) = \sqrt{x}$, what is expected utility?
A) $\sqrt{40} + \sqrt{90}$
B) $(\sqrt{40} + \sqrt{90})/2$
C) 65
D) $(40 + 90)/2$

12. If a risk-loving individual has $U(x) = x^2$, they will:
A) Always prefer a sure thing
B) Choose lower expected value gambles
C) Prefer risky lotteries with the same expected value
D) Refuse any uncertain option

13. Risk preference can be inferred from:
A) Income level alone
B) How utility changes with wealth
C) The slope of the budget line
D) The market price

14. A risk-averse person's utility curve:
A) Is convex to the origin
B) Is a straight line
C) Is concave to the origin
D) Always intersects the x-axis

15. A higher risk premium implies:
A) Lower risk aversion
B) More risk-loving behavior
C) Higher risk aversion
D) Certainty preference

16. Intertemporal choice refers to decisions involving:
A) Multiple people in a single time period
B) Consumption at different points in time
C) Price uncertainty
D) Budget maximization in one period

17. The discount factor reflects:
A) Interest rate
B) Time preference
C) Risk aversion
D) Consumer surplus

18. A lower discount factor implies:
A) The individual is more patient
B) The individual prefers future consumption
C) The individual is more impatient
D) Higher marginal utility

19. If the utility today is $U(C_1)$, and in the future is $\delta U(C_2)$, then the individual maximizes:
A) $C_1 + C_2$
B) $U(C_1) + U(C_2)$
C) $U(C_1) + \delta U(C_2)$
D) $\delta U(C_1) + U(C_2)$

20. Time inconsistency refers to:
A) Changing utility functions
B) Preferences shifting over time in a non-rational way
C) Always preferring future consumption
D) Ignoring time in choices

21. An exponential discounting model assumes:
A) Preferences are time inconsistent

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B) Constant rate of time preference
C) People always prefer current consumption
D) Future is valued more

22. A person who saves more today:
A) Has a low discount rate
B) Is highly risk-averse
C) Is very impatient
D) Values present consumption more

23. The regret criterion involves:
A) Choosing based on expected utility
B) Minimizing the worst possible loss
C) Minimizing the maximum regret from not choosing the best
D) Ignoring all outcomes

24. The regret criterion is best applied when:
A) Probabilities are known
B) Future outcomes are certain
C) There is no probability assigned to outcomes
D) There is no uncertainty

25. Under the regret criterion, a decision-maker evaluates:
A) Expected utility across all outcomes
B) The utility difference between the chosen and best possible outcome in each state
C) Certainty equivalents
D) Only the best possible outcome

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